

POINTS

30 points

SUBMISSION

File upload -- PDF

STEP

Step 3 of 3 -- Brake Booster Lab

Complete every field. Use ShopKey for specifications. Find your instructor for sign-off when this section is finished -- do not submit until initialed.

Vehicle Application

MAKE	MODEL	YEAR	ENGINE
TRANSMISSION	PROD. DATE	MODEL CODE	BUILD LOCATION

Battery OCV Test -- Perform before beginning hands-on steps. Record results below.

Battery Open Circuit Voltage

V Spec: 12.6V (fully charged)

OK/NG:

Technical Service Bulletins (Required)

TSBs Found:

List any TSBs found -- include number and title. If none found, write "none found." Search terms still required.

Search Terms Used:

Required -- list every search term used, even if no TSBs were found.

Step 1: Booster Location and Visual Inspection

3 PTS

Using the assigned vehicle, locate the brake booster. Using ShopKey, locate the inspection procedure and perform it.

Describe the brake booster location on the assigned vehicle:

Using ShopKey, locate the inspection procedure. Perform the procedure and record your findings:

Step 2: Vacuum Supply Test**2 PTS**

Connect a vacuum gauge to the vacuum supply hose at the brake booster. Start the engine and record vacuum at idle and at 2,000 RPM. Minimum spec: 18 in. Hg at idle.

Vacuum at idle		in. Hg	Spec: 18+ in. Hg minimum	OK/NG:
Vacuum at 2,000 RPM		in. Hg	Spec: Should equal or exceed idle reading	

Vacuum supply condition -- note any irregularities or low readings:

Step 3: Vacuum Depletion Test**4 PTS**

With engine OFF: Connect a vacuum gauge to the booster check valve port (between the check valve and booster). Push the brake pedal fully three times, recording the vacuum reading after each application. A booster in good condition should provide at least two full power-assisted applications.

Application	Vacuum Reading	Notes / Observations
Pedal Application 1		
Pedal Application 2		
Pedal Application 3		

How many full power-assisted brake applications did you get?

What does the depletion pattern tell you about the booster's vacuum reserve and check valve function?

Step 4: Booster Operation Check (Pedal Drop Test)**3 PTS**

Pump the brake pedal several times with the engine off to deplete stored vacuum. Apply firm pressure to the pedal and start the engine -- the pedal should drop/sink as vacuum builds.

Pedal feel with engine OFF (vacuum depleted):

Describe -- should feel hard/stiff with no vacuum assist.

Did the pedal drop when engine was started?

Spec: Yes -- drop indicates booster is functioning

Pedal feel with engine running:

Describe -- should feel lighter / assisted compared to engine off.

Describe what you observed and what it tells you about booster operation:

Step 5: One-Way Check Valve Inspection**4 PTS**

Remove the check valve from the booster. Inspect it visually, then perform the blow-through test: air should pass freely toward the engine side but NOT toward the booster side.

Describe the physical condition of the check valve (cracks, deformation, debris):

Blow toward booster side -- air passes?

Spec: Should NOT pass (no = correct)

OK/NG:

Blow toward engine side -- air passes?

Spec: Should pass freely (yes = correct)

OK/NG:

Check valve overall condition and action taken:

Step 6: Valve-Disconnected Booster Test

2 PTS

Leave the check valve disconnected from the booster but still connected to the vacuum hose. Repeat the pedal drop test from Step 4 and note any differences.

What changed compared to Step 4 with the check valve disconnected?

Explain why the check valve affects vacuum storage in the booster:

Reconnect the check valve before proceeding.

Step 7: Master Cylinder Pushrod Measurement

6 PTS

Using ShopKey, locate the specified pushrod protrusion length for the assigned vehicle. Measure the pushrod and compare to spec.

ShopKey pushrod protrusion specification:

SPECIFIED PUSHROD LENGTH / PROTRUSION	UNITS	SOURCE (SECTION / PAGE)

Measured pushrod protrusion		Spec: Compare to ShopKey spec	OK/NG:
-----------------------------	----------------------	-------------------------------	-----------------------------

Was pushrod adjustment required?

If adjustment was required, describe the correction performed:

Step 8: Customer Recommendation and Labor Time

4 PTS

Based on your inspection and test results, write a recommendation for the customer.
Use ShopKey to look up the labor time for brake booster removal and replacement (R&R).

Based on your findings, what is your recommendation to the customer?

ShopKey labor time for brake booster R&R:

LABOR TIME (HRS)	OPERATION DESCRIPTION

Return the vehicle to original condition. Any additional concerns noted during service?

INSTRUCTOR SIGN-OFF -- VEHICLE WALKTHROUGH (2 POINTS)

Review this section, initial below, and note any corrections. Student may not submit until initialed.

INITIALS:

DATE:

Earned: _____ / 2